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FSPP0034 PHYSICS I **TUTORIAL 2**

Chapter 2-4

1. To make a 0.40 μF capacitor, what area must the plates have if they are to be separated by a 2.8 mm air gap ?

   [Ans: 130 m^2]

2. An electric field of 4.80×10^5 V/m is desired between two parallel plates, each of area 21.0 cm^2 and separated by 0.25 cm of air. What charge must be on each plate?

   [Ans: 8.92×10^{-9} C]

3. Suppose in Fig.1 that $C_1=C_2=C_3 = 16.0$ μF and $C_4=28.5$ μF.
   If the charge on C_2 is 12.4 μC, determine :
   (i) the charge on each of the capacitors
   (ii) the voltage V_{ab}

   [Ans: $Q_1=Q_2=12.4$ μC; $Q_3=24.8$ μC; $Q_4=37.2$ μC
   $V_1=V_2=0.775$ V; $V_3= 1.55$ V ;$V_4=1.31$ V; $V_{ab}=2.86$ V]

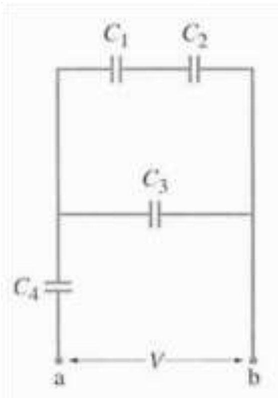


Figure 1

4. What is the capacitance of two square parallel plates 4.2 cm on a side that are separated by 1.8 mm of paraffin? [Ans:1.9×10^{-11} F]

5. What is the diameter of a 1.00 m length of tungsten wire whose resistance is 0.32 Ω?

   [Ans:4.7×10^{-4} m]

6. How much would you have to raise the temperature of copper wire (originally at 20 °C) to increase its resistance by 15 %? [Ans: 22 °C]

7. A 115 V fish-tank heater is rated at 95 W. Calculate:
   (a) the current through the heater when its operating
   (b) its resistance

   [Ans: (a) 0.83 A , (b) 140 Ω]

8. An ac voltage, whose peak value is 180 V, is across a 380 Ω resistor. What are the peak and rms currents in the resistor? [Ans: (a) 0.33 A , (b) 0.47 A]

9. The peak value of an alternating current in a 1500 W device is 5.4 A. What is the rms voltage across it? [Ans: 390 V]

10. A 0.65-mm-diameter copper wire carries a tiny current of 2.3 μA. Estimate:
    (a) the electron drift velocity, (b) the current density, and (c) the electric field in the wire.

    [Ans: (a) 5.1×10^{-10} ms^{-1}, (b) 6.9 Am^{-2}, (c) 1.2×10^{-17} Vm^{-1}]

5) $\rho = \frac{RA}{l}$

$5.6\times10^{-8} = \frac{0.32(\pi r^2)}{1}$

$r = 2.36\times10^{-4}$

$d = 2r = 4.72\times10^{-4}\ m$

6) $\rho_T = \rho_0(1+\alpha(T-T_0))$, $\alpha = 0.0068$

Let $\rho_0 = 1.68\times10^{-8}$, $\rho_T = 1.932\times10^{-8}$

$1.932\times10^{-8} = 1.68\times10^{-8}(1+0.0068(T-20))$

$T = 42.06°C$, $\Delta T = 42-20 = 22°C$

7) (a) $P = VI$

$95 = 115(I)$

$I = 0.826A$

(b) $V = IR$

$115 = 0.826(R)$

$R = 139.21\ \Omega$

8) $P = \frac{1}{2}I_0^2R$, $P = \frac{1}{2}\frac{V_0^2}{R}$

$\frac{180^2}{380} = I_0^2(380)$

$I_0 = 0.474A$

$I_{rms} = \frac{I_0}{\sqrt{2}}$

$= 0.335A$

9) $P = \frac{1}{2}I_0^2R$

$1500 = \frac{1}{2}(5.4)^2R$

$R = 102.88\ \Omega$

$P = \frac{1}{2}\times\frac{V_0^2}{R}$

$1500 = \frac{1}{2}\times\frac{V_0^2}{102.88}$

$V_0 = 555.55V$

$V_{rms} = \frac{V_0}{\sqrt{2}}$

$= \frac{555.55}{\sqrt{2}}$

$= 392.84V$

10) (a) $v = \frac{I}{nAq}$, $n = 8.5\times10^{28}e^-/m^3$

$= \frac{2.3\times10^{-6}}{8.5\times10^{28}(\pi\times0.325^2)(1.6\times10^{-19})}$

$= 5.1\times10^{-10}ms^{-1}$

(b) current density $= \frac{\text{current}}{\text{area}}$

$= \frac{2.3\times10^{-6}}{\pi\times0.325\times10^{-3\,2}}$

$= 6.93Am^{-2}$

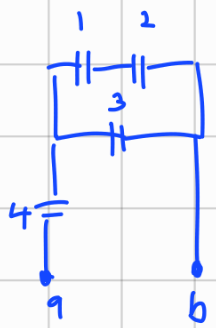
1) $C = 0.40\,\mu F$, $A = ?$, $d = 2.8\,mm$

$C = \frac{\varepsilon_0 A}{d}$, $0.40\times10^{-6} = \frac{8.85\times10^{-12}(A)}{2.8\times10^{-3}}$, $A = 126.55\,m^2$

2) $E = 4.80\times10^5$, $V = Ed$, $A = \frac{21cm^2}{1}\times\frac{0.01}{1cm}\times\frac{0.01}{1cm} = 2.1\times10^{-3}\,m^2$, $d = 2.5\times10^{-3}\,m$

$C = \frac{\varepsilon_0 A}{d} = 7.434\times10^{-12}F$, $V = 4.80\times10^5(2.5\times10^{-3}) = 1200\,V$

$Q = VC = 8.921\times10^{-9}\,C$

3) 

$\frac{1}{C_{12}} = \frac{1}{C_1} + \frac{1}{C_2}$

$C_{12} = 8\,\mu F$

$C_{12} + C_3 = C_x$

$C_x = 24\,\mu F$

$\frac{1}{C_y} = \frac{1}{C_x} + \frac{1}{C_4}$

$C_y = 13.03\,\mu F$

$Q_1 = Q_2 = Q_{12}$

$= 12.4\,\mu C$

$Q_3 = V_3 C_3$

$= 1.55(16)$

$= 24.8\,\mu C$

$Q_x = Q_4 = Q_{12} + Q_3$

$= 12.4 + 24.8$

$= 37.2\,\mu C$

$V_1 = V_2 = \frac{Q_1}{C_1}$

$= \frac{12.4}{16}$

$= 0.775V$

$V_{12} = 2(0.775)$

$= 1.55V$

$V_3 = V_{12}$

$= 1.55V$

$V_4 = \frac{Q_4}{C_4}$

$= \frac{37.2}{28.5}$

$= 1.31V$

$V_{ab} = \frac{Q}{C}$

$= \frac{37.2}{13.03}$

$= 2.85V$

4) $Area = 0.042\times0.042$

$= 1.764\times10^{-3}\,m^2$

$d = 1.8\,mm$

$= 1.8\times10^{-3}\,m$

$C = \frac{A\varepsilon_0}{d} = \frac{1.764\times10^{-3}\times8.85\times10^{12}}{1.8\times10^{-3}}$

$= 8.673\times10^{-12}$

11. Calculate the terminal voltage for a battery with an internal resistance of 0.90 Ω and an emf of 6.00 V when the battery is connected in series with 81 Ω resistor.

[Ans: (a) 5.93 V]

12. Determine (*a*) the equivalent resistance of the circuit shown in Fig. 2,
(*b*) the voltage across each resistor,

[Ans: (a) 1332 Ω , (b) V_{960}= 8.6 V, V_{820}= V_{680}= 3.4 V]

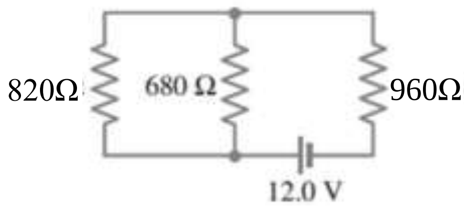


Figure 2

13. Calculate the currents in each resistor of Fig.3

[Ans: I_{120} = I_{82} = 0.24 A, I_{64} = 0.15 A, I_{25} = I_{110} = 0.092 A]

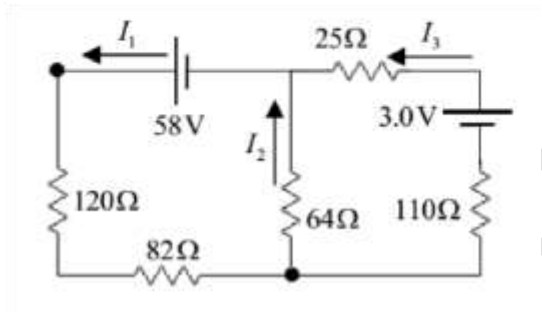


Figure 3

14. In Fig. 4 the total resistance is 15.0 kΩ and the battery's emf is 24.0 V. If the time constant is measured to be 24.0 μs, calculate :
(*a*) the total capacitance of the circuit and
(*b*) the time it takes for the voltage across the resistor to reach 16.0 V after the switch is closed.

[Ans: (a) C= 1.60 x 10^{-9} F (b) t = 9.73 μs]

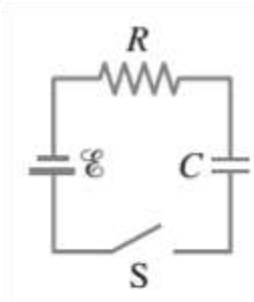


Figure 4

11) $\varepsilon = I(R+r)$

$6 = I(81+0.9)$

$I = 0.073A$

$V = IR$

$= 0.073(81)$

$= 5.93V$

12) (a) $R_1 = 820$, $R_2 = 680$, $R_3 = 960$

$\frac{1}{R_{12}} = \frac{1}{820} + \frac{1}{680}$

$= 371.73\,\Omega$

$R_{eff} = R_{12} + R_3$

$= 371.73 + 960$

$= 1331.73\,\Omega$

(b) $V = IR$

$12 = I(1332)$

$I = 0.009A$

$I = I_{12} = I_3$

$V_3 = 0.009(960)$

$= 8.64V$

$V_{12} = I_{12}R_{12} = V_1 = V_2 = 0.009(372)$

$= 3.348V$

13) $I_1 = I_2 + I_3$ — (1)

$58 = 202I_1 + 64I_2$ — (2)

$3 = 135I_3 - 64I_2$ — (3)

$61 = 202I_1 + 135I_3$

$61 = 202(I_2 + I_3) + 135I_3$

$61 = 202I_2 + 337I_3$

$3 = -64I_2 + 135I_3$

$183 = 606I_2 + 1011I_3$

$(-)\ 183 = -3904I_2 + 8235I_3$

$0 = 4510I_2 - 7224I_3$

$4510I_2 = 7224I_3$

$I_2 = \frac{7224}{4510}I_3$

$3 = 135I_3 - 64\left(\frac{7224}{4510}\right)I_3$

$I_3 = \frac{3}{32.5}$

$= 0.092A$

$I_2 = \frac{7224 \times 0.092}{4510}$

$= 0.15A$

$I_1 = 0.242A$

14) (a) $\tau = RC$

$C = \frac{\tau}{R} = \frac{24\times10^{-6}}{15000}$

$= 1.6\times10^{-9}F$

(b) $V = V_0(1 - e^{-\frac{t}{RC}})$

$16 = 24 - 24e^{-\frac{t}{RC}}$

$-8 = -24e^{-\frac{t}{RC}}$

$\frac{1}{3} = e^{-\frac{t}{RC}}$

$\ln(\frac{1}{3}) = -\frac{t}{24\mu}\ln(e)$